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***B.Tech. Degree III Semester Supplementary Examination
April 2018***

**ME 15-1305 METALLURGY AND MATERIAL SCIENCE
(2015 Scheme)**

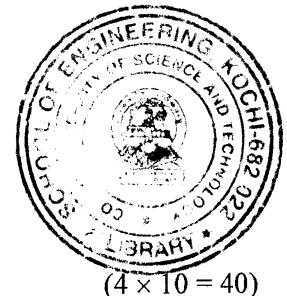
Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Draw the following crystallographic planes in a BCC unit cell and list the position of the atoms whose centers are intersected by each of the planes (100), (110) and (111).
 (b) Define the homogenous nucleation process for the solidification of pure metal.
 (c) Differentiate between twist and twin boundary.
 (d) What are the conditions favourable for extensive solid solubility of one element in another?
 (e) What is the normalizing heat treatment for steel? What are some of its purpose?
 (f) Explain the significance of Jominy end-quench test.
 (g) Distinguish between anelastic and visco elastic behavior.
 (h) Comment on the mechanism of creep.
 (i) What are the four basic types of cast iron?
 (j) Explain the importance and applications of shape memory alloys.



PART B

(4 × 10 = 40)

- II. (a) Calculate the atomic packing factor for the FCC structure.
 (b) Describe and illustrate edge and screw type of dislocations.
 (c) What do you mean by polymorphism?

OR

- III. (a) Distinguish between equiaxed and columnar grains in a solidified metal structure.
 (b) Distinguish between steady and non steady diffusion process.
 (c) What are the effects of point defects on the properties of materials?
- IV. (a) Explain the different types of solid solution.
 (b) Draw the Fe-C diagram and explain the salient features.
 (c) Discuss briefly the electro plating technique.

OR

- V. (a) Distinguish between Martempering and Austempering process.
 (b) Briefly discuss the flame and induction hardening processes.
 (c) Write the equation of Gibbs phase rule and define each of the terms.
- VI. (a) Describe the plastic deformation mechanism of a single crystal.
 (b) What necessitates recovery and recrystallisation processes? How do they influence the properties of materials?
 (c) Distinguish between work hardening and grain size strengthening.

OR

(P.T.O.)

- VII. (a) Describe Griffith's theory of brittle fracture.
(b) Discuss the mechanism of fatigue failure. Suggest the methods to improve fatigue life of materials.
(c) How is the ductility of a metal normally affected by cold and hot working? Explain.
- VIII. (a) What are some of the properties of alloy steels that make them important?
(b) Name few light metal alloys and discuss their engineering applications.
(c) Discuss the mechanical and thermal properties of ceramics.
- OR**
- IX. (a) Discuss the compositions, properties and uses of some copper alloys.
(b) What are some practical applications of composite materials?
(c) Comment on the technological significance of nano materials.

